

팍리스 실무형 일경험 프로그램

SDV 아키텍처 기반 자율주행 통합 시스템 구현 실습



4일차

06 D3-G + VCP-G 통합 CAN 실습 (IPC
APP)

Telechips  **TOPST**

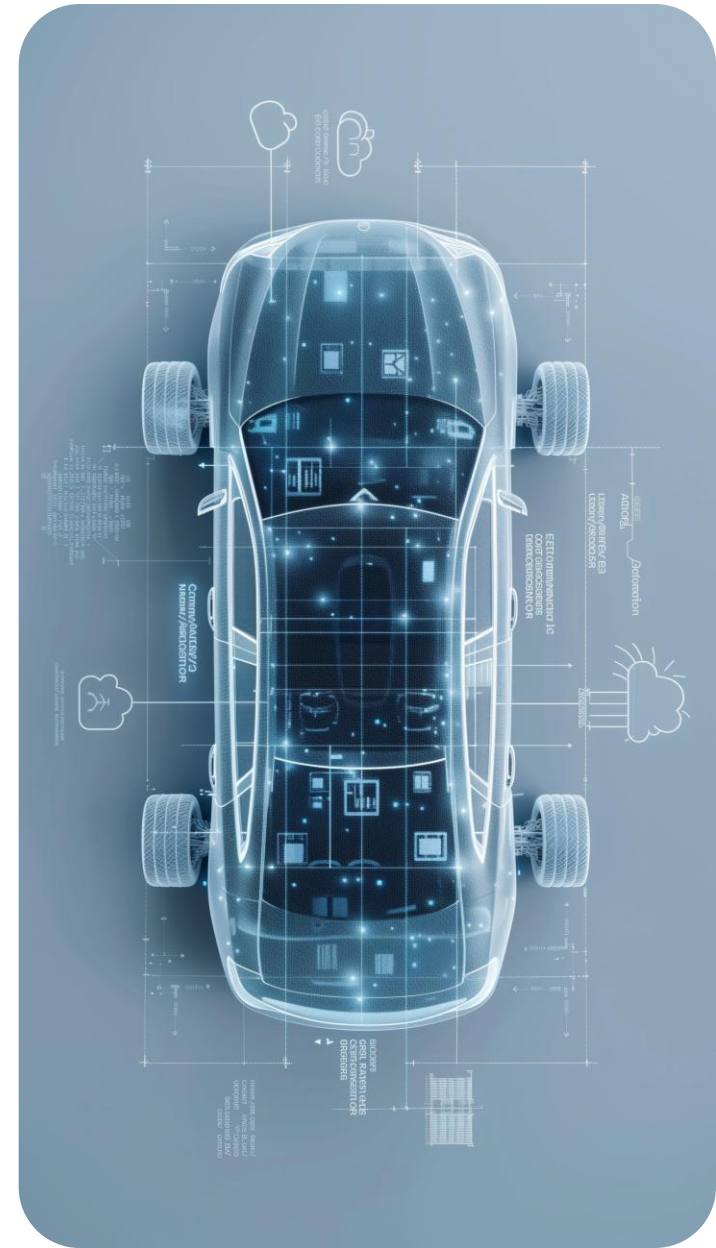


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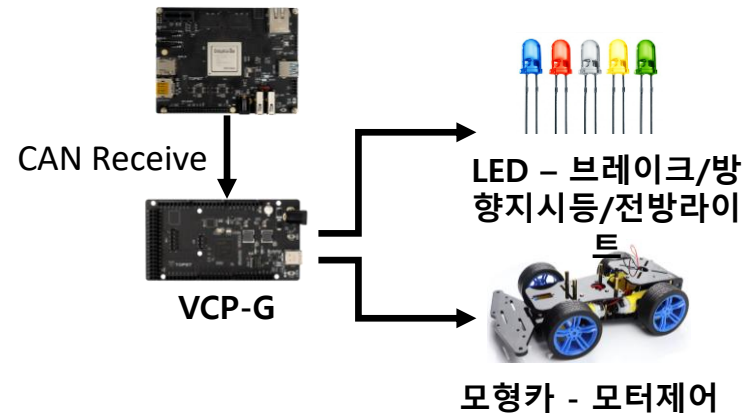
CAN 통신 보드 종합 실습(D3-G+VCP-G) 02



Control Zone 개요

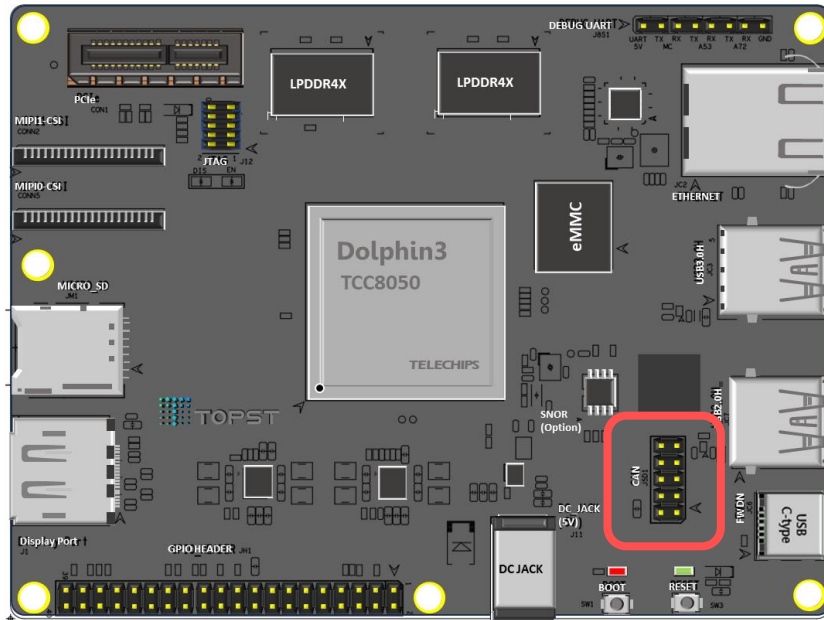
Q) TOPST 보드는 어떤 형태로 CAN 통신을 지원하는지

Q) 두 보드간 CAN 통신을 지원하기 위해선 물리적인 연결만 필요?

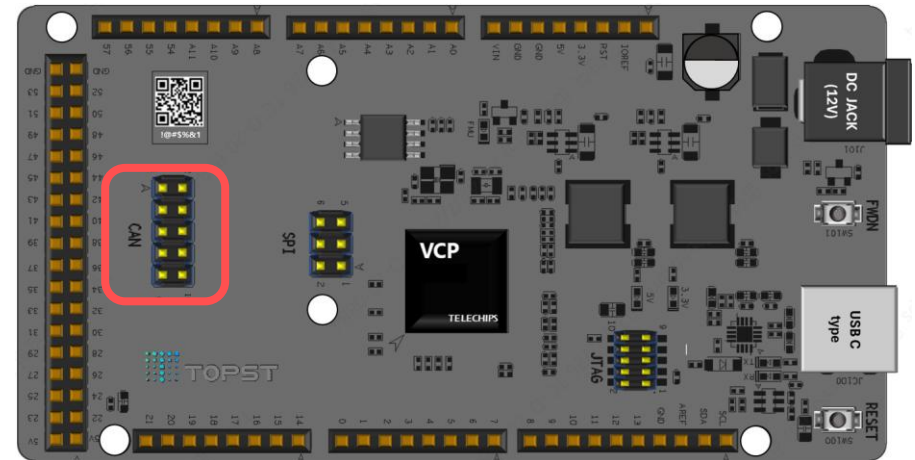


Support to CAN pin

- Q) TOPST 보드는 어떤 형태로 CAN을 지원하는지
- TOPST D3-G CAN 지원은 3ch까지 가능 – 10개의 PIN 형태로 지원
- TOPST VCP-G CAN 지원은 3ch까지 가능 – 10개의 PIN 형태로 지원



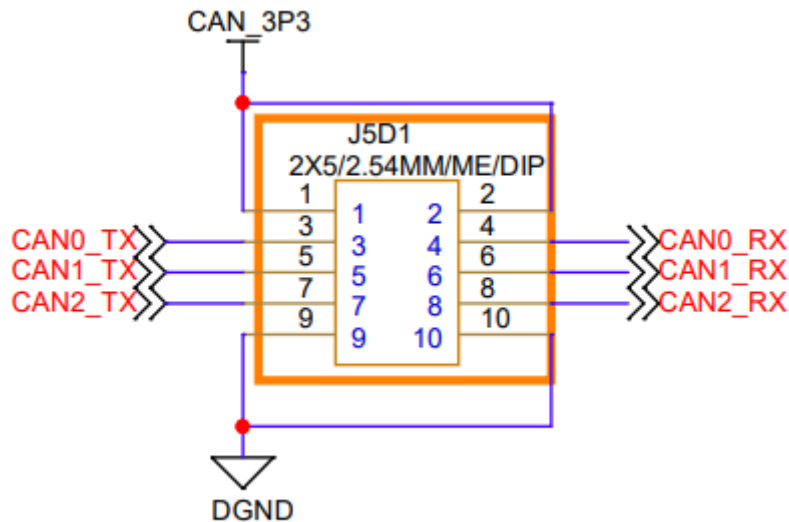
D3-G



VCP-G

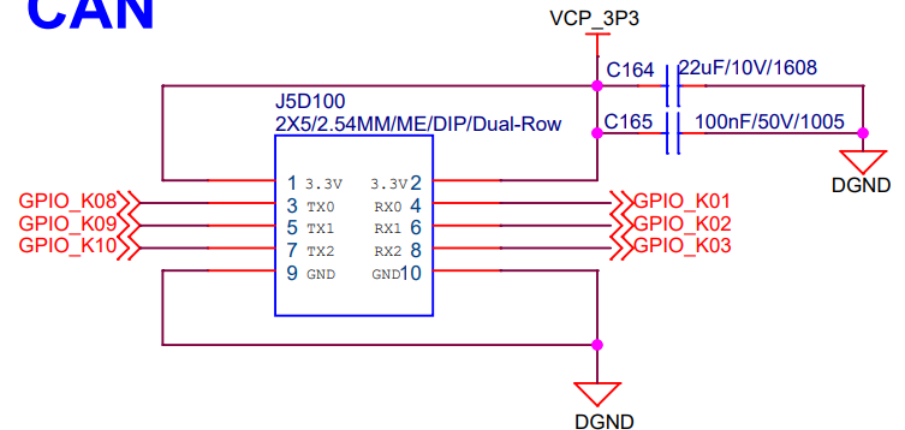
Support to CAN pin

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- 핀의 형태는 두 보드 동일



D3-G CAN Port

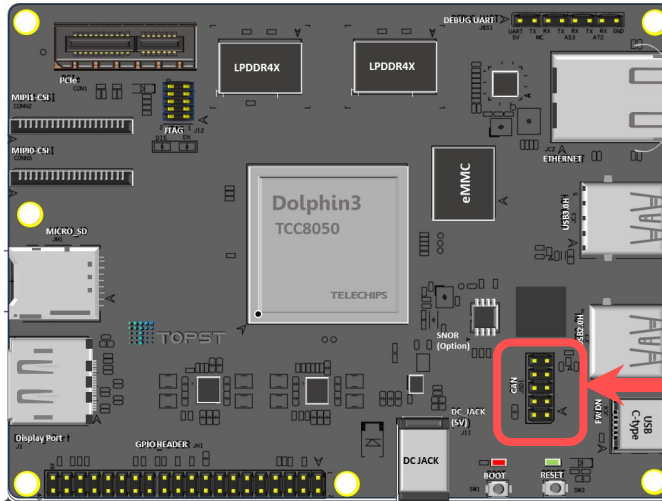
CAN



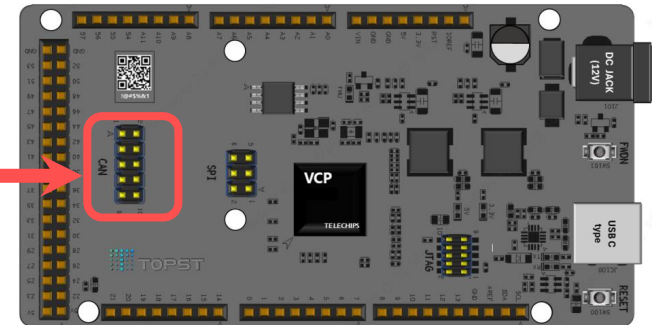
VCP-G CAN Port

Support to CAN pin

- Q) 물리적으로 서로 연결하기만 한다면 CAN통신이 되나요?
 - CAN 버스는 차동 신호(CAN High, CAN Low)로 통신하기 때문에 단순히 TX, RX 방식의 통신을 할 수 없음
- D3-G와 VCP-G의 CAN 통신을 위한 CAN Transceiver 장비를 사용
- 본 과정에서 SN65HVD230 VP230 CAN 모듈 사용 예정



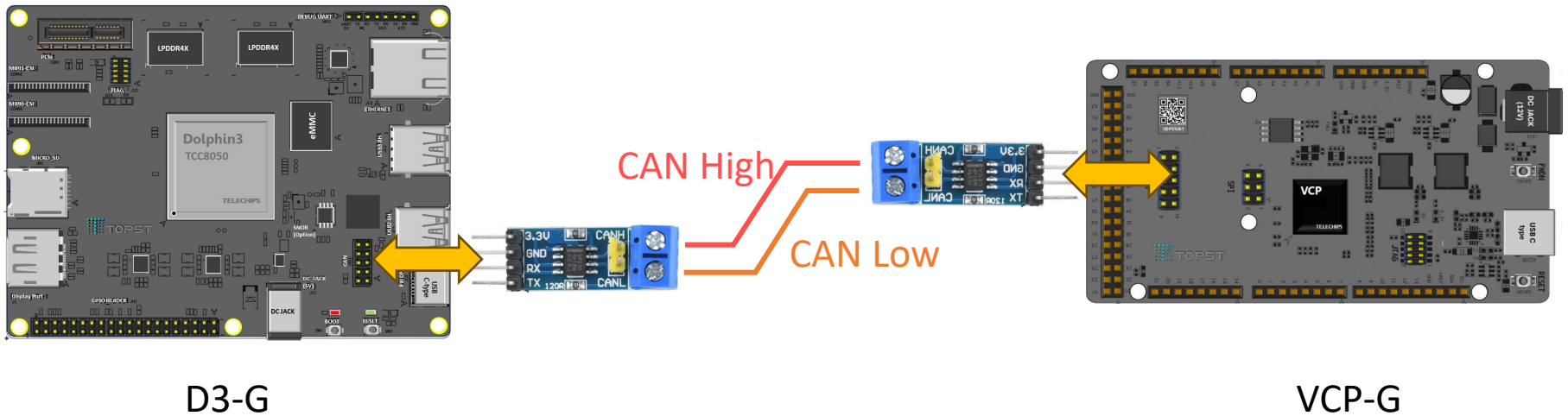
D3-G



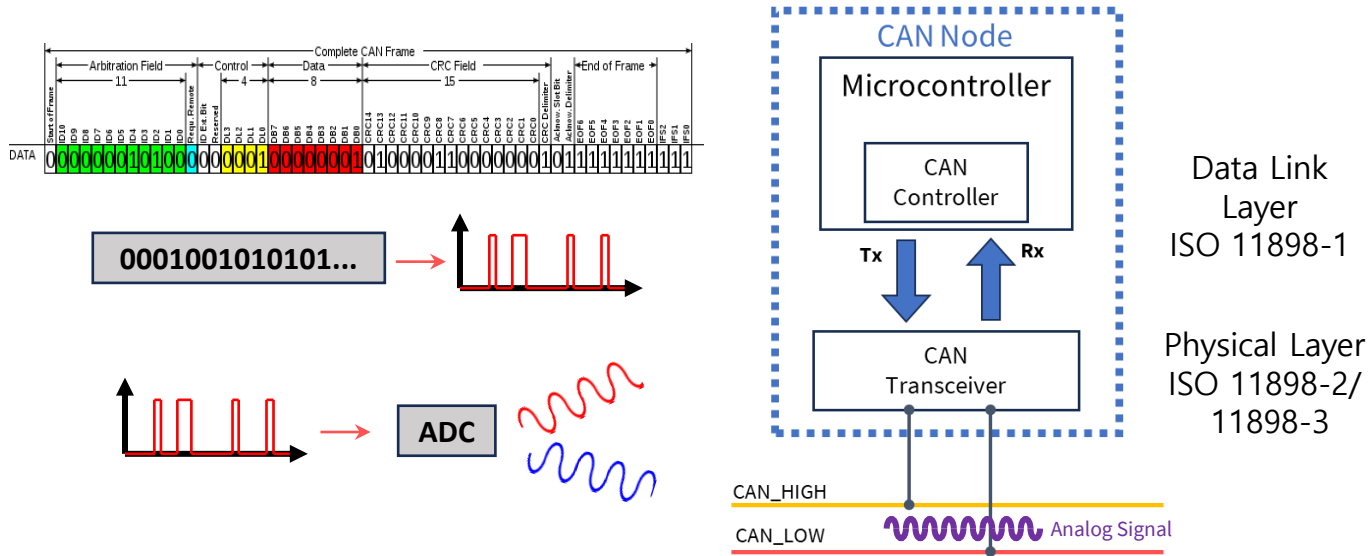
VCP-G

Support to CAN pin

- 따라서, 별도의 CAN Transceiver 장비를 사용해야, 보드간 CAN 통신을 수행할 수 있다.



CAN Message 전송 흐름



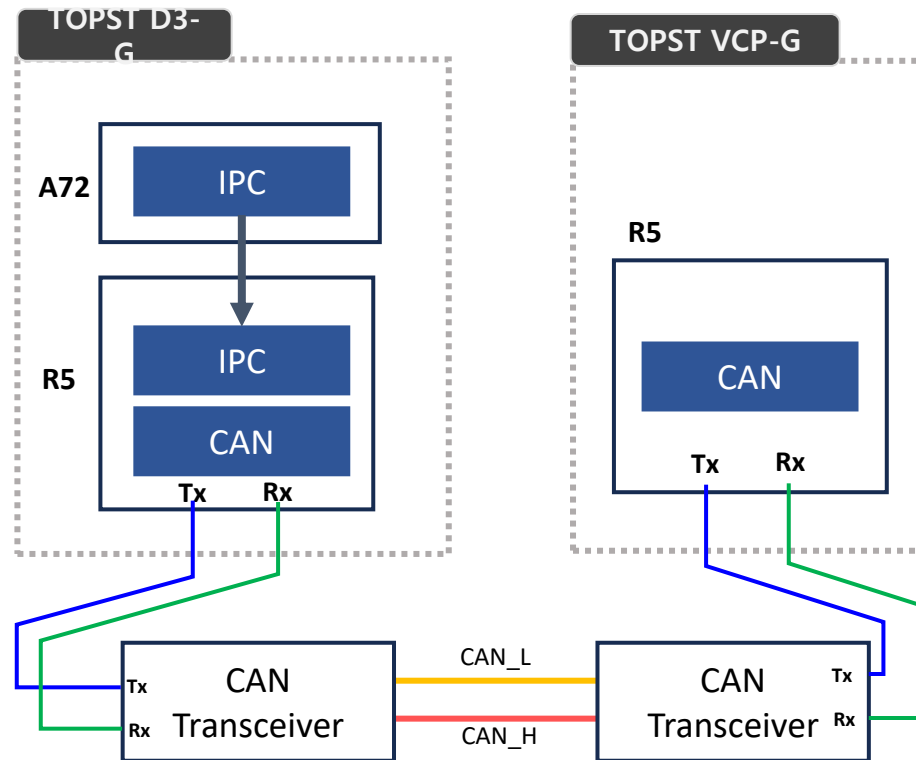
ISO 11898-1: CAN 데이터 링크 계층을 정의한다. CAN 프로토콜에서 데이터의 전송 및 오류 처리 기능 뿐만 아니라 표준 CAN 메시지의 구조와 프레임의 형식을 규정한다.

ISO 11898-2: 고속 CAN(CAN High-Speed)의 물리 계층을 정의한다. 차량 내 통신에서 주로 사용되며, 데이터 전송 속도는 최대 1 Mbps이다.

ISO 11898-3: 저속 CAN(CAN Low-Speed)의 물리 계층을 정의한다. 주로 차량의 간단한 네트워크 및 저속 통신에 사용되며 Fault Tolerant(내고장성, 결함 허용) 특징을 가지고 있다.

- CAN 통신과 연관된 동작을 하는 장치 : CAN controller
 - CAN controller 는 MCU 내부에 마련되어 있음
 - CAN 통신의 프로토콜을 따라서 실제로 CAN 통신을 할 수 있도록 지원
- CAN 컨트롤러에서 링크 계층 표준을 만족하며, 데이터의 전송, 수신 및 프로토콜을 담당한다.
- CAN Transceiver(**T**ransmitter + **R**eceiver)
 - 디지털 신호,아날로그 신호를 서로 변환하고 송수신하는 역할을 지원한다.
 - Controller에서 생성한 신호(메시지)를 ADC를 통해서 전압값으로 변환
 - 변환된 전압 신호(Analog Signal)을 CAN bus로 전송

TOPST 보드간 CAN 메시지 전송 흐름

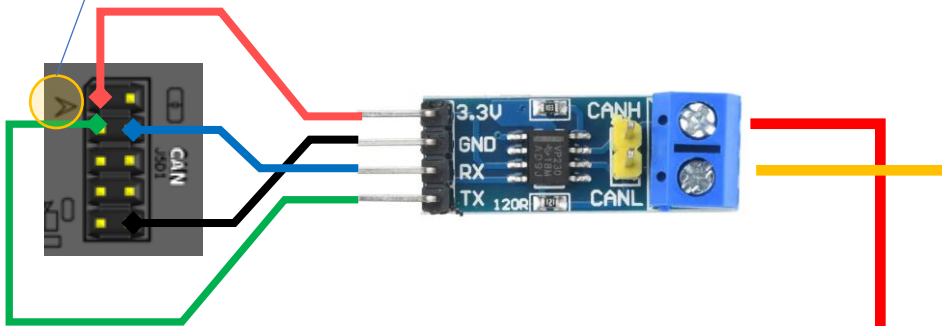


- 따라서 위와 같은 구조로 실습 수행
- D3-G [A72 >> R5] -> VCP-G 순으로 데이터 전송

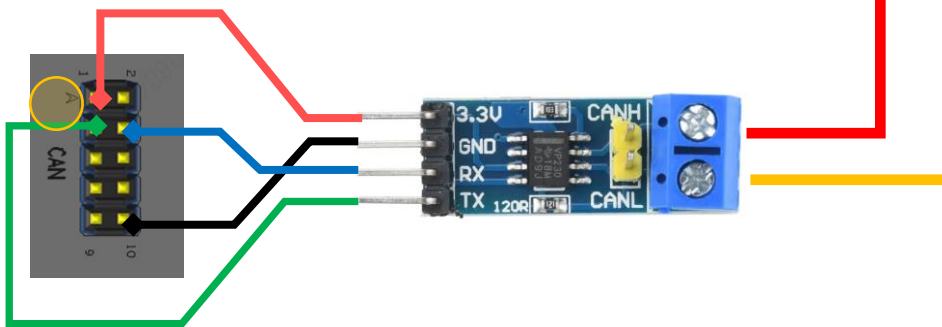
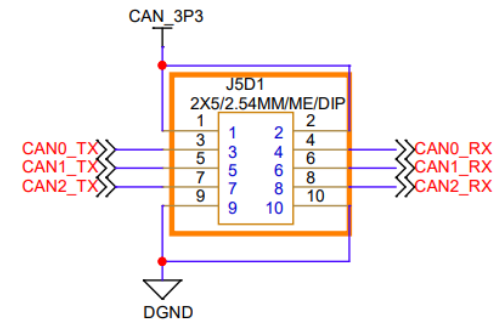
실습을 위한 보드 연결

두 보드간 연결을 위해 아래와 같이 회로를 결선한다.

화살표 표시 부분이 1번핀

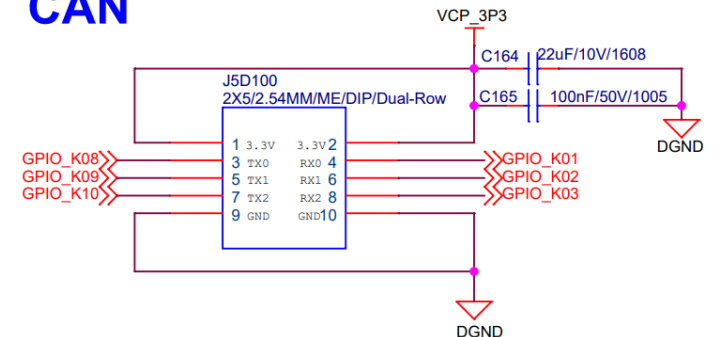


D3-G



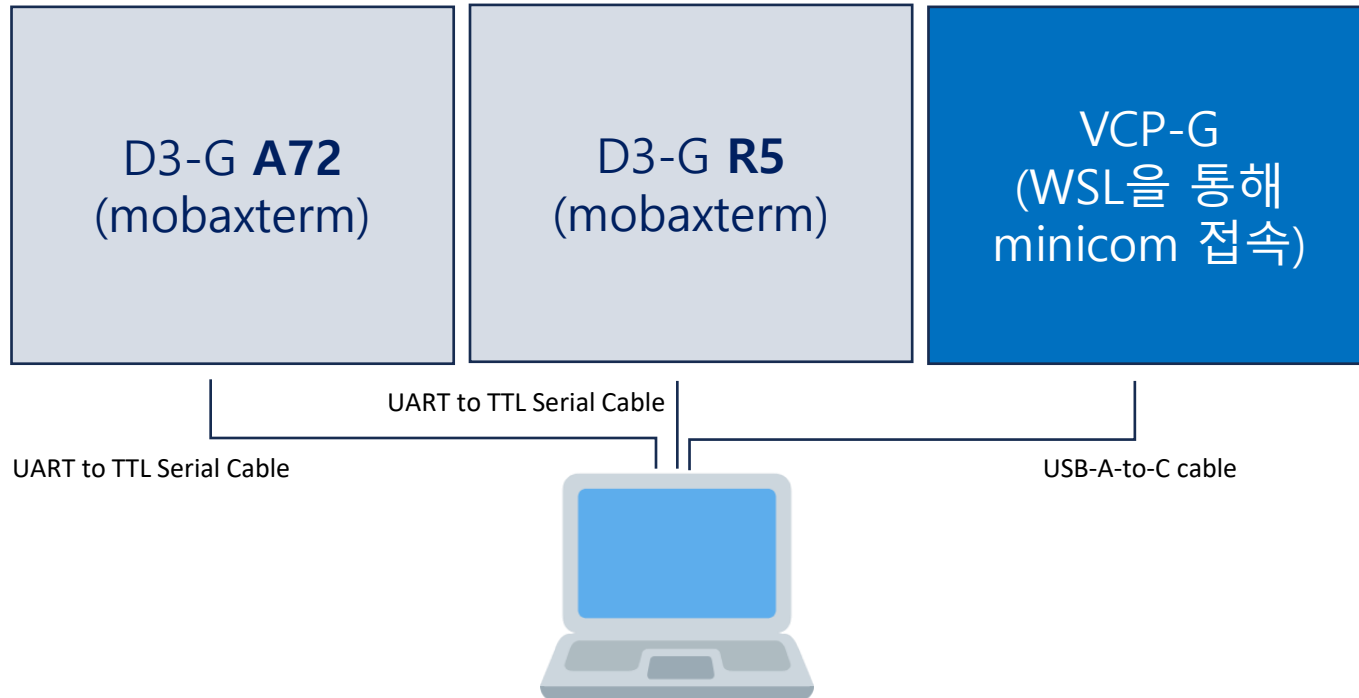
VCP-G

CAN



실습환경 세팅

- D3-G 보드와 VCP-G 보드와 UART 통신 세팅을 진행한다.



- A72 터미널내에서 IPC 명령어를 통해 R5 터미널과 VCP-G 터미널 화면을 관찰할 수 있다.
 - `$ cd ~/Education/d3-g\ app/`

실습환경 세팅

- A72 IPC_Example 실행 커맨드 리스트 예제

No.	Scenario	IPC Command
1	Brake Signal	<pre>\$ sudo python3 IPC_Example.py snd --canID 0x101 --sndDataHex 01 \$ sudo python3 IPC_Example.py snd --canID 0x101 --sndDataHex 02</pre>
2	Direction Signal	<pre>\$ sudo python3 IPC_Example.py snd --canID 0x102 --sndDataHex 0101 \$ sudo python3 IPC_Example.py snd --canID 0x102 --sndDataHex 0102 \$ sudo python3 IPC_Example.py snd --canID 0x102 --sndDataHex 0201 \$ sudo python3 IPC_Example.py snd --canID 0x102 --sndDataHex 0202</pre>
3	Emergency Signal	<pre>\$ sudo python3 IPC_Example.py snd --canID 0x103 --sndDataHex 01</pre>
4	HeadLight Signal	<pre>\$ sudo python3 IPC_Example.py snd --canID 0x104 --sndDataHex 01 \$ sudo python3 IPC_Example.py snd --canID 0x104 --sndDataHex 02</pre>
5	Fuel Status Signal	<pre>\$ sudo python3 IPC_Example.py snd --canID 0x105 --sndDataHex 33</pre>
6	Motor Speed Signal	<pre>\$ sudo python3 IPC_Example.py snd --canID 0x106 --sndDataHex 33</pre>
7	Motor Direction Signal	<pre>\$ sudo python3 IPC_Example.py snd --canID 0x107 --sndDataHex 7F</pre>

실습 결과

```

Serial (COM)
topst@TOPST:~/Education/d3-g app$ sudo python3 IPC_Example.py snd --canID
[sudo] password for topst:
Sorry, try again.
[sudo] password for topst:
args.mode: snd, args.channel: 1, args.txOnly: 0, args.canID:
0x0101
channel_bitmask: 0x01 (channels: 0)
tx_only_channel_bitmask: 0x00 (channels: None)
canID: 0x0101
sndData: b'\x01'
sndDataHex: 01
sndDataLength: 1
example : python3 IPC_Example.py sndrecv --sndData [data]
topst@TOPST:~/Education/d3-g app$ sudo python3 IPC_Example.py snd --canID
0x101 --sndDataHex 01
args.mode: snd, args.channel: 1, args.txOnly: 0, args.canID: 0x0101
channel_bitmask: 0x01 (channels: 0)
tx_only channel bitmask: 0x00 (channels: None)
canID: 0x0101
sndData: b'\x01'
sndDataHex: 01
sndDataLength: 1
example : python3 IPC_Example.py sndrecv --sndData [data]
topst@TOPST:~/Education/d3-g app$

(D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
(D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
(D) [CAN ] [CAN_Edu_SetSendingData:1567]
ch_bitmask: 0x1, tx_only_bitmask: 0x0, can_id: 0x101
(D) [CAN ] [CAN_Edu_SetSendingData:1583] [CH] : 0x0, [ID] : (D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
: 0x101, [DATA SIZE] : 1 (5), [DATA] :
(D) [CAN ] [CAN_Edu_SetSendingData:1586] 0x01
(D) [CAN ] [CAN_Edu_SetSendingData:1591]
(D) [CAN ] [CAN_Edu_SetSendingData:1599] gNowMetaInfo.channel : 0x7
(D) [CAN ] [CAN_Edu_DemoSend:1332] canTxMsg => id: 257, isFE: 0, extID: 0, len: 1, data:
(D) [CAN ] [CAN_Edu_DemoSend:1345]
ret: 0
(D) [CAN ] [CAN_Edu_DemoSend:1369]
CAN_SEND ch: 0 id: 257 mDataLength:1 (1:10)
(D) [CAN ] [CAN_Edu_DemoSend:1375] Successful transmission.
(D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
(D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
(D) [CAN ] [CAN_Edu_SetSendingData:1567]
ch_bitmask: 0x1, tx_only_bitmask: 0x0, can_id: 0x101
(D) [CAN ] [CAN_Edu_SetSendingData:1583] [CH] : 0x0, [ID] : (D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
: 0x101, [DATA SIZE] : 1 (5), [DATA] :
(D) [CAN ] [CAN_Edu_SetSendingData:1586] 0x01
(D) [CAN ] [CAN_Edu_SetSendingData:1591]
(D) [CAN ] [CAN_Edu_SetSendingData:1599] gNowMetaInfo.channel : 0x7
(D) [CAN ] [CAN_Edu_DemoSend:1332] canTxMsg => id: 257, isFE: 0, extID: 0, len: 1, data:
(D) [CAN ] [CAN_Edu_DemoSend:1345]
ret: 0
(D) [CAN ] [CAN_Edu_DemoSend:1369]
CAN_SEND ch: 0 id: 257 mDataLength:1 (1:10)
(D) [CAN ] [CAN_Edu_DemoSend:1375] Successful transmission.

[CAN DEMO]
< Channel 0 Received Message Information >
*****
[ID] : 0x101, [DATA SIZE] : 1, [DATA] :
0x01
*****
Brake Light ON
[CAN DEMO] No message
  
```

D3-G A72 (IPC_Example)



D3-G R5



VCP-G

실습 결과

The image displays three terminal windows showing the results of CAN communication tests. The first window (left) shows the command prompt for the D3-G A72 board, where the user runs a Python script to send data. The second window (middle) shows the command prompt for the D3-G R5 board, which receives the data and sends a response. The third window (right) shows the command prompt for the VCP-G board, which also receives the data and sends a response. Red boxes highlight specific data points in each window.

```

(D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
(D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
(D) [CAN ] [CAN_Edu_SetSendingData:1567]
ch_bitmask: 0x1, tx_only_bitmask: 0x0, can_id: 0x101
(D) [CAN ] [CAN_Edu_SetSendingData:1583] [CH] : 0x0, [ID] : (D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
0x101, [DATA SIZE] : 1 (5), [DATA] :
(D) [CAN ] [CAN_Edu_SetSendingData:1586] 0x02
(D) [CAN ] [CAN_Edu_SetSendingData:1591]
(D) [CAN ] [CAN_Edu_SetSendingData:1599] gNowMetaInfo.channel : 0x7
(D) [CAN ] [CAN_Edu_DemoSend:1332] canTxMsg => id: 257, isFE: 0, extID: 0, len: 1, data:
(D) [CAN ] [CAN_Edu_DemoSend:1345]
ret: 0
(D) [CAN ] [CAN_Edu_DemoSend:1369]
CAN_SEND ch: 0 id: 257 mDataLength:1 (1:10)
(D) [CAN ] [CAN_Edu_DemoSend:1375] Successful transmission.
(D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
(D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
(D) [CAN ] [CAN_Edu_SetSendingData:1567]
ch_bitmask: 0x1, tx_only_bitmask: 0x0, can_id: 0x101
(D) [CAN ] [CAN_Edu_SetSendingData:1583] [CH] : 0x0, [ID] : (D) [MBOX ] [MBOX_DevQget:1314] empty for ch[3] siDevId[2]
: 0x101, [DATA SIZE] : 1 (5), [DATA] :
(D) [CAN ] [CAN_Edu_SetSendingData:1586] 0x02
(D) [CAN ] [CAN_Edu_SetSendingData:1591]
(D) [CAN ] [CAN_Edu_SetSendingData:1599] gNowMetaInfo.channel : 0x7
(D) [CAN ] [CAN_Edu_DemoSend:1332] canTxMsg => id: 257, isFE: 0, extID: 0, len: 1, data:
(D) [CAN ] [CAN_Edu_DemoSend:1345]
ret: 0
(D) [CAN ] [CAN_Edu_DemoSend:1369]
CAN_SEND ch: 0 id: 257 mDataLength:1 (1:10)
(D) [CAN ] [CAN_Edu_DemoSend:1375] Successful transmission.
  
```

D3-G A72 (IPC_Example)



D3-G R5



VCP-G

실습 결과

```

topst@TOPST:~/Education/d3-g app$ sudo python3 IPC_Example.py snd
--canID 0x103 --sndDataHex 01
args.mode: snd, args.channel: 1, args.txOnly: 0, args.canID: 0x0103
channel_bitmask: 0x01 (channels: 0)
tx_only channel bitmask: 0x00 (channels: None)
canID: 0x0103
sndData: b'\x01'
sndDataHex: 01
sndDataLength: 1
example : python3 IPC_Example.py sndrecv --sndData [data]
topst@TOPST:~/Education/d3-g app$

(D)[CAN ][CAN_Edu_SetSendingData:1567]
ch_bitmask: 0x1, tx_only_bitmask: 0x0, can_id: 0x101

(D)[CAN ][CAN_Edu_SetSendingData:1583] [CH] : 0x0, [ID] (D)[MBOX ][MBOX_Dev
Qget:1314] empty for ch[3] siDevId[2]

: 0x101, [DATA SIZE] : 1 (5), [DATA] :

(D)[CAN ][CAN_Edu_SetSendingData:1586] 0x02
(D)[CAN ][CAN_Edu_SetSendingData:1591]

(D)[CAN ][CAN_Edu_SetSendingData:1599] gNowMetaInfo.channel : 0x7

(D)[CAN ][CAN_Edu_DemoSend:1332] canTxMsg => id: 257, isFE: 0, extID: 0, le
n: 1, data:
(D)[CAN ][CAN_Edu_DemoSend:1345]
ret: 0

(D)[CAN ][CAN_Edu_DemoSend:1369]
CAN_SEND ch: 0 id: 257 mDataLength:1 (1:10)

(D)[CAN ][CAN_Edu_DemoSend:1375] Successful transmission.

(D)[TIMER][TIMER_Disable:594] Channel (1) is disabled

[PMIO][PMIO_STR_SetTimer:1727] pmio Set Timer disable, reason clear
[PMIO][PMIO_STR_TimeoutInterruptHandler:1750] Timeout : ACC ON signal
[PMIO][PMIO_STR_TimeoutInterruptHandler:1751] Request : STR mode
(D)[MBOX ][MBOX_DevQget:1314] empty for ch[3] siDevId[2]

(D)[MBOX ][MBOX_DevQget:1314] empty for ch[3] siDevId[2]

(D)[CAN ][CAN_Edu_SetSendingData:1567]
ch_bitmask: 0x1, tx_only_bitmask: 0x0, can_id: 0x103

(D)[CAN ][CAN_Edu_SetSendingData:1583] [CH] : 0x0, [ID] (D)[MBOX ][MBOX_Dev
Qget:1314] empty for ch[3] siDevId[2]

: 0x103, [DATA SIZE] : 1 (5), [DATA] :

(D)[CAN ][CAN_Edu_SetSendingData:1586] 0x01
(D)[CAN ][CAN_Edu_SetSendingData:1591]

[CAN DEMO]
< Channel 0 Received Message Information >
*****
[ID] : 0x101, [DATA SIZE] : 1, [DATA] :
0x01
*****

Brake Light ON
[CAN DEMO] No message
[CAN DEMO]

< Channel 0 Received Message Information >
*****
[ID] : 0x101, [DATA SIZE] : 1, [DATA] :
0x02
*****

Brake Light OFF
[CAN DEMO] No message
[CAN DEMO]

< Channel 0 Received Message Information >
*****
[ID] : 0x101, [DATA SIZE] : 1, [DATA] :
0x02
*****

Brake Light OFF
[CAN DEMO] No message
[CAN DEMO]

< Channel 0 Received Message Information >
*****
[ID] : 0x103, [DATA SIZE] : 1, [DATA] :
0x01
*****

Emergency Signal ON<-->OFF
[ControlBreadBoardSensors][204] undefined can id !!!
[CAN DEMO] No message
  
```

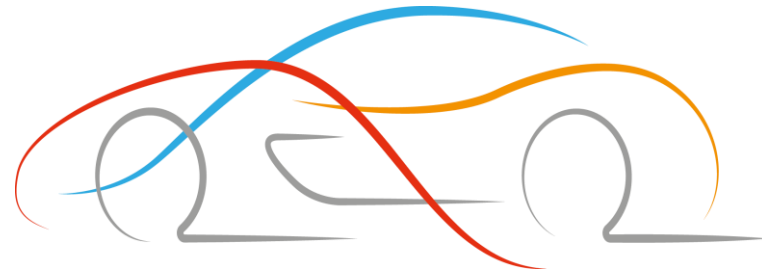
D3-G A72 (IPC_Example)



D3-G R5



VCP-G



Thank You